COMP202 Week 3 - solutions

1. A biased coin comes up heads with probability p, where $0 < p < 1$. The coin is flipped repeatedly until the first head appears.

   Let X denote the number of flips made (including the final flip that gives heads).

   Compute $\mathbb{E}[X]$.

   Solution: This solution is similar to the analysis we did in class for the case $p = 1/2$. Consider the first coin flip.

   - With probability p, the flip is heads and the experiment ends after one flip.
   - With probability $1 - p$, the flip is tails; this costs one flip, and we are now in exactly the same situation as at the start.

   Therefore,
   $$\mathbb{E}[X] = p \cdot 1 + (1 - p) \cdot (1 + \mathbb{E}[X]).$$

   Simplifying,
   $$\mathbb{E}[X] = 1 + (1 - p)\mathbb{E}[X],$$

   so
   $$p\,\mathbb{E}[X] = 1,$$

   and hence
   $$\mathbb{E}[X] = \frac{1}{p}.$$

   Intuition: If heads is unlikely (that is, p is small), we expect to wait a long time before seeing one. If $p = 1/2$, we recover the familiar value $\mathbb{E}[X] = 2$. If p is close to 1, the expected number of flips is close to 1, which matches our intuition.

2. Two fair six-sided dice are rolled once. Let $D = |X - Y|$ be the absolute difference between the two outcomes.

   Compute $\mathbb{E}[D]$.

Solution: You might be tempted to use linearity of expectation somehow but unfortunately this doesn't work when absolute values are involved. So you just need to do this from first principles.

There are 36 equally likely outcomes (X, Y) with $X, Y \in \{1, 2, 3, 4, 5, 6\}$.

For $d = 0$, we must have $X = Y$. There are 6 such outcomes, so

$$\Pr(D = 0) = \frac{6}{36}.$$

For $d \geq 1$, the number of outcomes with $|X - Y| = d$ is $2(6 - d)$, justified below. Choose the smaller value $s \in \{1, \ldots, 6 - d\}$ (note that the smaller value cannot be larger than $6-d$, since otherwise the larger value would be larger than 6, which is impossible) and set the larger value to $s + d$. This gives $6 - d$ outcomes with $X < Y$ and the same number with $X > Y$.

Hence,

$$\Pr(D = d) = \frac{2(6 - d)}{36} \quad \text{for } d = 1, 2, 3, 4, 5.$$

Explicitly, the number of outcomes for each value of D is:

d	0	1	2	3	4	5
#	6	10	8	6	4	2

which sums to 36, as expected.

The expected value of D is

$$\mathbb{E}[D] = \sum_{d=0}^{5} d \Pr(D = d) = \frac{0 \cdot 6 + 1 \cdot 10 + 2 \cdot 8 + 3 \cdot 6 + 4 \cdot 4 + 5 \cdot 2}{36}.$$

Evaluating the numerator gives 70, and hence

$$\mathbb{E}[D] = \frac{70}{36} = \frac{35}{18}.$$

3. Bob has a set A of n nuts and a set B of n bolts, such that each nut in A has a unique matching bolt in B. Unfortunately, the nuts in A all look the same, and the bolts in B all look the same as well. The only kind of comparison that Bob can make is to take a nut-bolt pair (a, b) (such that $a \in A$ and $b \in B$) and test them to see if the threads of a are larger, smaller, or a perfect match with the threads of b.

   Describe an efficient algorithm for Bob to match up all of the nuts and bolts. What is the running time of this algorithm, in terms of the number of nut-bolt comparisons that Bob must make?

Solution: You should be used to this by now. Apply the randomized QuickSort algorithm in the following manner.

Pick a nut (at random), use this to sort the bolts into two sets, those having diameters smaller than the nut and larger than the nut, finding the matching bolt in the process. (You can repeat this process if you like, until you have the two sets being "not too big" as in the normal randomized QuickSort routine.)

Now that you have found one matching nut/bolt pair, use the bolt to sort the nuts into two piles, those with threads larger or smaller than the bolt. Now you have two pile of nuts and two piles of bolts. note that since each bolt has a matching nut, and vice-versa (and the matches are unique), then the pile of bolts with the "smaller" threads is equal in size to the pile of nuts with the "smaller" threads, and the same is true of the other pile of nuts and bolts.

Then you repeat this process on each of the matching piles of nuts and bolts.

Using a version of the randomized QuickSort procedure to do this, how many comparisons will you have to make?

If we use the randomized procedure to choose our "pivots" to ensure the piles aren't too big, then, in expectation since it's a randomized procedure, we'll be doing at most $O(n \log n)$ comparisons. The first round takes $n + n$ comparisons (well, it's really $n + (n - 1)$), so there's $O(n)$ comparisons there.

The second round we'll do at most $O(n)$ comparisons again. (Recall, the randomization procedure ensures that each subset is size at most $\frac{3}{4}n$. So we would do at most $2\frac{3}{4}n$ comparisons to subdivide the matching set of nuts and bolts into two smaller piles. thus overall we'll do at most n comparisons again).

The number of rounds until we're done will be (in expectation) at most $\log_{\frac{4}{3}} n$, as in the case of the usual randomized QuickSort procedure. So overall we'll have at most (in expectation) $O(n \log n)$ comparisons.

4. You are given an unsorted array A of n integers and an integer T.

   Describe an efficient algorithm to determine whether the *median* (that is, the $n/2$'th largest element) of A is strictly greater than T.

   Solution: As should hopefully be clear, you can directly use Randomized Select to get the $n/2$'th largest number in expected $O(n)$ queries, and just check if that number is larger than T.

   However, there's a much simpler solution to this problem that just requires a linear scan: maintain a counter that counts the number of items seen that are larger than T. Initialize the counter to 0, and simply scan the list from left to right, increasing the counter by 1 whenever you see an element larger than T. If this counter exceeds $n/2$ answer yes, else answer no.

The key difference that makes this problem so easy in contrast with Selection is that we're given the number T in this problem.